COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/08/2026	**DATE SUBMITTED:**	07/09/2026
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ACTIVITY NO. 1.1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

What procedures did you perform?

Create an Algorithm, Pseudocode, Flowchart, and C++ Program for the following:

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm

- Gets the scores of Quiz 1, Quiz 2, and Quiz 3.
- After getting the scores of Quiz 1, 2, and 3 get the sum of all quizzes
- After getting the sum of Quiz 1, 2 and 3 divide to get the average of all quizzes

- Display the Total sum of all quizzes added up.
- Display the Average of all Quizzes after dividing it by 3.

Pseudocode

```
START

DECLARE score1, score2, score3, sum
DECLARE avg

DISPLAY "Please Enter Quiz 1 Score: "
GET score1
DISPLAY "Please Enter Quiz 2 Score: "
GET score2
DISPLAY "Please Enter Quiz 3 Score: "
GET score3

GET sum
  ADD score1 + score2 + score3

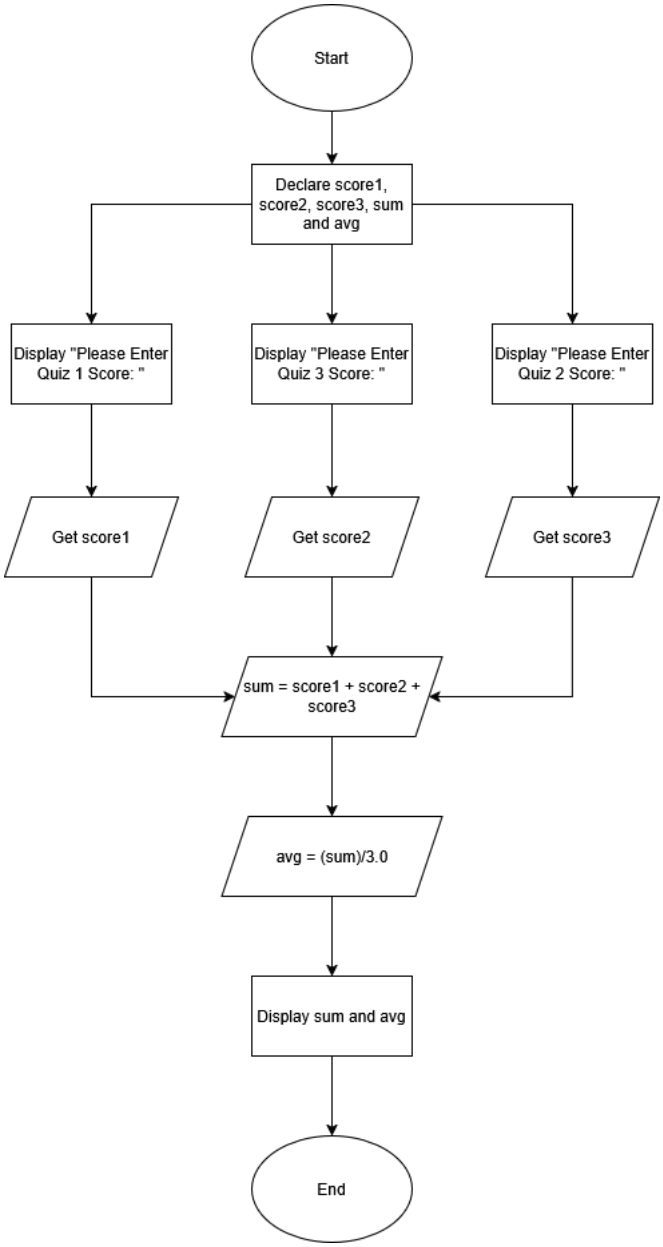
GET avg
 GET sum
   DIVIDE (sum)/3.0
```

DISPLAY "The Total for all Quizzes is: "
DISPLAY sum

DISPLAY "The Average Grade for all Quizzes is: "
DISPLAY avg

END

Flowchart



Solution to Task 1

For task one, I first started by declaring score1, score2, score3, and sum as integers as there are no decimal points in individual quiz scoring, for the avg however I decided to use float instead in order to tally up and get the most accurate result possible by including decimal points. Then it asks the user to type/input 3 different quiz scores, then once it achieves all 3 quiz scores it adds it all together and gets the total sum.

Then after getting the sum it gets the average of all 3 quiz scores by using the value in the sum integer and divides it by 3.0. Then after it meets all the conditions it displays the total sum of all quiz scores and the average.

Screenshot of the Code & it's Output

```cpp
// Astudillo, Thomas Joseph T.
// CPE21S2

#include <iostream>
using namespace std;

int main() {
    int score1, score2, score3, sum;
    float avg;

    cout << "Please Enter Quiz 1 Score: " << endl;
    cin >> score1;
    cout << "Please Enter Quiz 2 Score: " << endl;
    cin >> score2;
    cout << "Please Enter Quiz 3 Score: " << endl;
    cin >> score3;

    sum = score1 + score2+ score3;
    avg = sum/3.0;


    cout << "The Total for all Quizzes is: " << sum << endl;
    cout << "The Average Grade for all Quizzes is: " << avg << endl;

    return 0;
}
```

```
PS C:\Users\TIPQC\Downloads\CPE21S2_Seatwork 1.1_ASTUDILLO-20260710T024054Z-2-001\CPE21S2_Seatwork 1.1_ASTUDILLO>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAd
PS C:\Users\TIPQC\Downloads\CPE21S2_Seatwork 1.1_ASTUDILLO-20260710T024054Z-2-001\CPE21S2_Seatwork 1.1_ASTUDILLO>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAd
pters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-dd2iw0iz.rgp' '--stdout=Microsoft-MIEngine-Out-0lkdm1m1.3ge' '--stderr=Microsoft-MIEngine-Error-azydairc.kuy' '--pid=Microsoft-M
Engine-Pid-wuabkoke.cmd' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Please Enter Quiz 1 Score:
Please Enter Quiz 1 Score:
15
Please Enter Quiz 2 Score:
23
Please Enter Quiz 3 Score:
18
The Total for all Quizzes is: 56
The Average Grade for all Quizzes is: 18.6667
```

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm

- Gets the name of item1, item2, and item3
- Gets the prices for item1, item2, and item 3 using price1, price2, and price3
- Displays the items you have named and its total amount after adding it all up using totalitems and totalprice

- It then asks for the payment for the items
- If payment is >= the total price it then prints out "Payment Successful!"
- It then gives you the change depending on the totalprice of the items and payment using the formula: payment - totalprice
- It then displays all the items and their respective price within the Transaction Receipt.
- It then terminates the session.

- If the payment is < the total price however, it prints out "Insufficient amount."
- It then terminates the session.

Pseudocode

```
START

DECLARE item1, item2, item3
DELCARE price1, price2, price3, totalprice
DECLARE payment, change

DISPLAY "Please Enter Item No. 1: "
GET item1
DISPLAY "Please enter The Price of Item No. 1: "
GET price1
DISPLAY "Please Enter Item No.2: "
GET item2
DISPLAY "Please enter The Price of Item No. 2: "
GET price2
DISPLAY "Please Enter Item No.3: "
GET item3
DISPLAY "Please enter The Price of Item No. 3: "
GET price3


DISPLAY item1, item2, item3
DISPLAY price1, price2, price3
GET totalprice
totalprice = price1 + price2 + price3
DISPLAY "The total amount is: "
        DISPLAY totalprice

DISPLAY "How much would you want to pay?: "
GET payment

IF payment >= totalprice
```

```
DISPLAY "Payment Successful!"
GET change
change = payment - totalprice
DISPLAY "Your total change is: "
        DISPLAY change

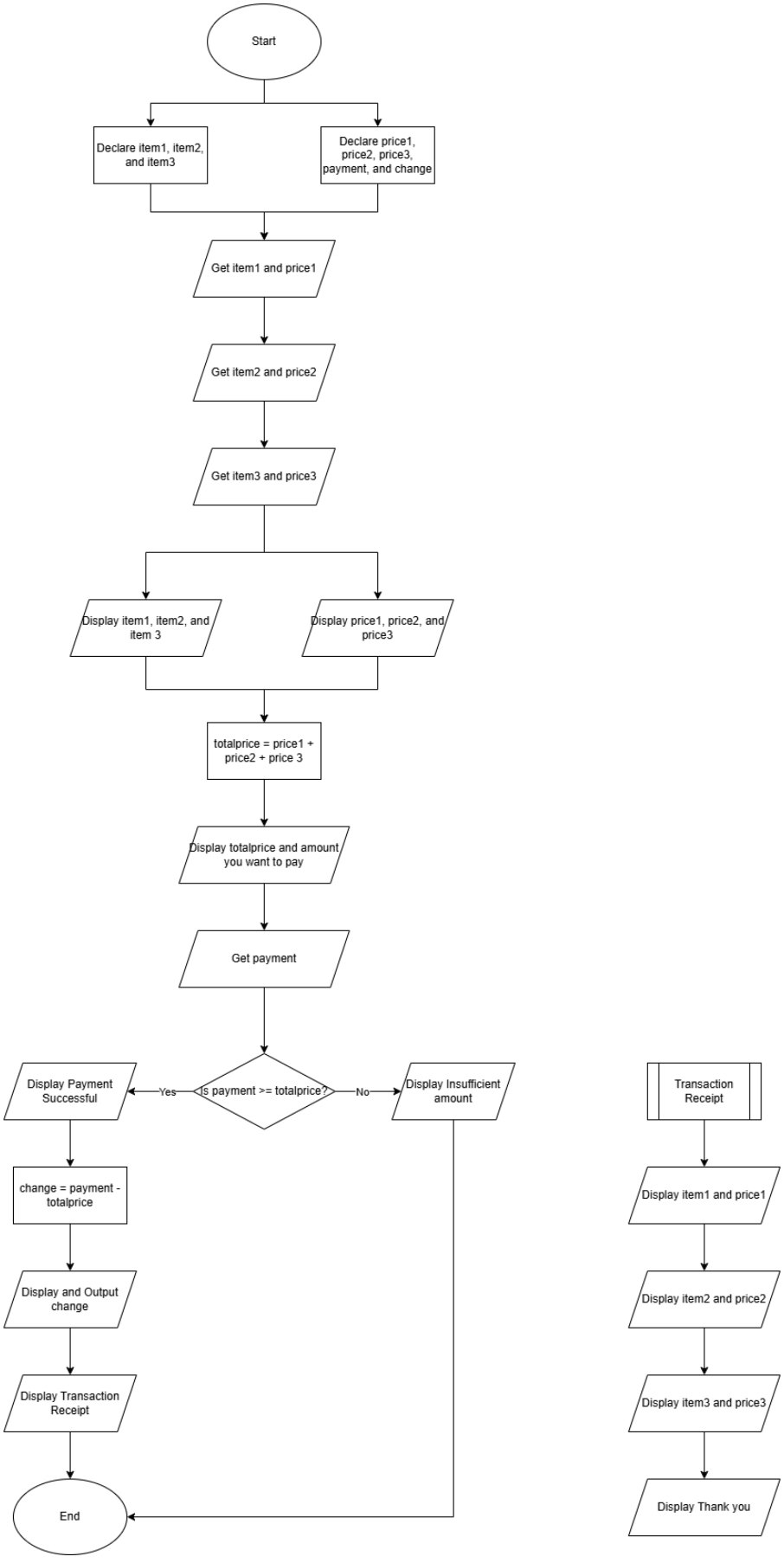
DISPLAY "Transaction Receipt"
DISPLAY item1 and price1
DISPLAY item2 and price2
DISPLAY item3 and price3
DISPLAY "Thank you for shopping!"

END

ELSE
DISPLAY "Insufficient amount."

END
```

Flowchart


Start
Declare item1, item2, and item3
Declare price1, price2, price3, payment, and change
Get item1 and price1
Get item2 and price2
Get item3 and price3
Display item1, item2, and item 3
Display price1, price2, and price3
totalprice = price1 + price2 + price 3
Display totalprice and amount you want to pay
Get payment
Is payment >= totalprice?
Yes
No
Display Payment Successful
Display Insufficient amount
change = payment - totalprice
Display and Output change
Display Transaction Receipt
End
Transaction Receipt
Display item1 and price1
Display item2 and price2
Display item3 and price3
Display Thank you

Solution to Task 2

For Task 2, I started by declaring item1, item2, and item3 as String variables and price1, price2, price3, payment, and change as float variables. This is because item1 - item3 are the text data that contain no number value while price1 - price3 as well as payment and change can contain number values. Afterwards the terminal/program asks for 3 item names as well as their price, after this condition is fulfilled it then displays all items and their price respectively before displaying the totalprice, which is calculated using the formula "total price = price1 + price2 + price3". Then it asks for payment for the items, this is where i used if-else conditions.

If the payment is >= the total price it accepts the payment and gives the change which is calculated using payment - totalprice. After the change is generated it displays the transaction receipt which includes all of the items and their respective price along with a thank you message then terminates the session.

However, if the payment is < the total price it simply displays "Insufficient amount." and terminates the session.

Screenshot of the Code & it's Output

```
1   // Astudillo, Thomas Joseph T.
2   // CPE21S2
3
4   #include <iostream>
5   #include <string>
6   using namespace std;
7
8   int main() {
9       string item1, item2, item3;
10      float price1, price2, price3, totalprice;
11      float payment, change;
12
13      cout << "Please Enter Item No. 1: " << endl;
14      cin >> item1;
15      cout << "Please enter The Price of Item No. 1: " << endl;
16      cin >> price1;
17
18      cout << "Please Enter Item No. 2: " << endl;
19      cin >> item2;
20      cout << "Please enter The Price of Item No. 2: " << endl;
21      cin >> price2;
22
23      cout << "Please Enter Item No. 3: " << endl;
24      cin >> item3;
25      cout << "Please enter The Price of Item No. 3: " << endl;
26      cin >> price3;
```

```cpp
    cout << "The items you have are: " << item1 << "," << item2 << "," << item3 << "." << endl;
    cout << "The Price for the respective items are: " << price1 << "," << price2 << "," << price3 << "." << endl;
    totalprice = price1 + price2 + price3;
    cout << "The total amount is: " << totalprice << endl;
    cout << "How much would you want to pay?: " << endl;
    cin >> payment;

    if (payment >= totalprice) {
        cout << "Payment Successful!" << endl;
        change = payment - totalprice;
        cout << "Your total change is: " << change << endl;

        cout << "Transaction Receipt" << endl;
        cout << item1 << " : " << price1 << endl;
        cout << item2 << " : " << price2 << endl;
        cout << item3 << " : " << price3 << endl;
        cout << "Thank you for shopping!";

    } else {
        cout << "Insufficient amount.";
    }


    return 0;
}
```

```
PS C:\Users\TIPQC\Downloads\CPE21S2_Seatwork 1.1_ASTUDILLO-20260710T024054Z-2-001\CPE21S2_Seatwork 1.1_ASTUDILLO> & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-birjfupb.05x' '--stdout=Microsoft-MIEngine-Out-rtjssp00.42c' '--stderr=Microsoft-MIEngine-Error-sm2urzuu.45y' '--pid=Microsoft-MIEngine-Pid-5f2becug.gp4' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Please Enter Item No. 1:
Chicken
Please enter The Price of Item No. 1:
100
Please Enter Item No. 2:
Eggs
Please enter The Price of Item No. 2:
150
Please Enter Item No. 3:
Milk
Please enter The Price of Item No. 3:
89
The items you have are: Chicken,Eggs,Milk.
The Price for the respective items are: 100,150,89.
The total amount is: 339
How much would you want to pay?:
500
Payment Successful!
Your total change is: 161
Transaction Receipt
Chicken : 100
Eggs : 150
Milk : 89
Thank you for shopping!
PS C:\Users\TIPQC\Downloads\CPE21S2_Seatwork 1.1_ASTUDILLO-20260710T024054Z-2-001\CPE21S2_Seatwork 1.1_ASTUDILLO>
```

“I affirm that I will not give or receive any unauthorized help on this activity and that all work will be my own.”

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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